

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech. (EE) Examination

May 2014

EE-409 Digital Signal Processing

Date: 10.05.2014, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Draw & explain the basic Building Block of the Digital Signal Processing system. [07]

Q - 2 (a) Find whether the signals are power, energy or neither energy nor power signals and Determine the values of power and energy of the following signals. [06]

a) $x[n] = [1/3]^n u[n]$ **b)** $x[n] = \cos[(\pi/4)n]$ **c)** $x[n] = e^{j(\frac{\pi}{2}n + \frac{\pi}{8})}$

(b) For **Three** system mentioned below, Check whether they are time variant or time invariant, Static or dynamic & causal or non-causal [06]

(a) $y[n] = x[n]x[n-1]$ **(b)** $y[n] = x[n]u[n]$ **(c)** $y[n] = \log_{10}|x[n]|$

(c) Explain 2 distinct properties of z transform. [02]

OR

(a) State equation of a_0 , a_n & b_n for Continuous Fourier Transform. [06]
Also derive the equation of a_0 .

(b) Find the z transform & ROC of [04]

(a) $x[n] = 0.5^n u[-n]$ **(b)** $x[n] = a^{|n|}, |a| < 1$.

(c) Explain the concept of Negative Frequency shown on Frequency Spectrum plot. [04]

Q – 3 (a) Prove the following Properties of Discrete Fourier Transform. [06]

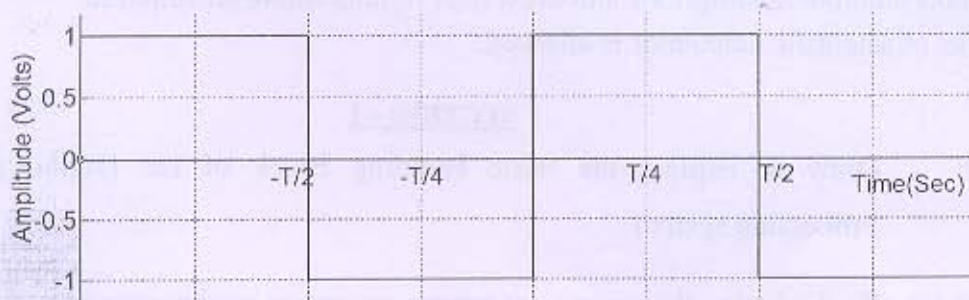
1. Linearity:

$$\text{DFT}[a_1 x_1(n) + a_2 x_2(n)] = a_1 X_1(K) + a_2 X_2(K).$$

2. Multiplication in Time:

$$\text{DFT}[x_1(n) \cdot x_2(n)] = (1/N) [X_1(K) * X_2(K)]$$

(b) Find out Fourier Series Co-efficients a_0 , a_n , b_n for following signal [08]
(from $-T/2$ to $T/2$ only).



OR

Q – 3 (a) Calculate Linear Convolution of following. [06]

$$x(n) = \{3, 1, 2, -1\}, \quad h(n) = \{3, 2, 1\}$$

(b) Explain any Four distinct application of Digital Signal Processing. [08]

SECTION – II

Q - 4 Discuss various types of Filter with it's Frequency Response Characteristics. [07]

Q -5 (a) Draw Signal Flow Graph for 8 point sequence using Decimation in Time (DIT) Algorithm. [07]

(b) Design a Low Pass Butterworth Filter using Bi-Linear Transformation, [07]
for satisfying the following constraints (Solve up to $H(s)$ only, $H(z)$ is not required):

Pass Band: 0-400 Hz.

Stop Band: 2.1 KHz to 4 KHz.

Pass Band Ripple: 2 dB.

Sampling Frequency: 10 KHz.

OR

(a) Derive Impulse Response equation $h_d(n)$ for a Low Pass Finite Impulse Response (FIR) Filter. [07]

(b) Draw Signal Flow Graph for 8 point sequence using Decimation in Frequency (DIF) Algorithm. [07]

Q – 6 (a) Design a Low Pass FIR Filter for $N=5$ & $\omega_c = 0.5$ radian for Hamming as well as Hanning Window. Also draw the signal flow graph for the derived $H(z)$. [08]

(b) Explain the Reasons with necessary calculations of following for 8 point FFT calculated by Decimation In Time (DIT) algorithm: [06]

1. Why $G(0)=G(4)$

2. Why $W_8^0 = -W_8^4$

OR

(a) Write short note on **Impulse invariant Transformation** method with it's S-Z Pole Mapping Diagram. [07]

(b) State equations required for Realization of **IIR** Filter using **Direct Form-I** and **Direct Form-II** method. Also draw the signal flow graph for both. [05]

(c) Realize following FIR system using Signal Flow Graph. [2]

$$H(z) = 0.2 + 5z^{-1} + 7z^{-2} + 3z^{-3}$$
